SYMBOL_TO_STATUS_CODE, 226
symbolic debuggers, 311
system testing, *see also* end-to-end testing
  cables, 239
  defined, 152
  Devise, 267
  Minitest, 253
  performance, 161, 254
  replacing controller tests with in Rails 5, xv, 58, 223
  RSpec, 152, 160–161
  running, 254
  simulating user interaction, 152–160
  vs. unit testing, 161
  when to use, 160–161

T

-t option, 323
tables, Cucumber step definitions with data tables, 178
--tag option, 320, 323
tags
  Capybara anchor tags, 156–157
  Cucumber, 292, 323
  RSpec, 320
tags method, 293
TDD (test-driven development), *see also* design and TDD; outside-in testing
  vs. acceptance testing, 5
  administrator and public access, 273
  advantages, xiii, 1–3
  associations and validations, 90
  costs and value, 68–70
  damage, 8
  defined, 3
  legacy code, 349–352, 359
  limitations, 7
  metaprocess diagram, 80
  process, 3
  rapid feedback, 75, 332
  rules, 8
  starting points, 14, 21, 80
  test structure, 30
  test-next coding, 7
  tests as documentation, 6, 82
  unit testing JavaScript example, 197–201
  uses, 5–7
  using first test to point to next test, 24
  vs. verification testing, 35
  view testing and, 233
teardown
  bypassing Rails in Minitest, 337
  examples and example groups, 20
  multiple, 245
  relative dates, 126
  test independence, 75
templates
  mailers, 232
  older controller tests, 228
  stubbing partial views, 235
test coverage and legacy code, 346, 348, 359
test data, *see* data
test doubles, *see also* fakes; mock objects; spies; stubs
  about, 8
  adapters, 294
  annotating expectations, 144–147
  balancing value and cost, 69
  bypassing Rails for speed, 338–341
  creating, 132–136, 143, 205
  creating in testdouble.js, 205
  defined, 129–132
  disadvantages, 132, 143
  finders and, 92
  full doubles, 132–134, 137, 249
  integration testing, 141
  Minitest with Mocha, 242, 249–252
  null objects, 133
  partial doubles, 132, 134–136, 204, 250
  performance, 143, 343
  removing dependencies in legacy code, 354–356
  for repeatability, 73
  request tests, 224–225
  seams, 356–359
  specifying behavior with, 141–144, 206–210
  terms for, 130
  tips, 147–149
  unit testing JavaScript with testdouble.js, 192, 204–213
  uses, 129
  verifying doubles, 133, 135, 205, 210–213, 249
  when to use, 147
test-first coding, *see* TDD (test-driven development)
test-next mode, 7
test.log file, 245
Test::Unit, 241
test_helper.rb file
  authentication and authorization with Devise, 263
  fixtures, 246
  Guard, 330
  Minitest, 242–243, 245
  Mocha, 249
  running single test, 322
  VCR, 291
testdouble package, 194
testdouble-jasmine package, 194
testdouble.js, 192, 194, 204–213
testers, software, 6
testing, *see also* acceptance testing; integration testing; Minitest; model testing; route testing; RSpec; system testing; TDD (test-driven development); test doubles; unit testing
  balancing cost and value, 68–70
  black-box testing, 349
  breaking multiple, 266
  cables, 239
  characterization tests, 350–352
  clarity, 67, 70–73
  focus on behavior, 29
  independence, 74
  jobs, 239, 253
  mailers, 229–233
  pending tests, 45, 48
  pyramid, 160
  repeatability, 73
  route testing, 218, 227, 259
  running tests on specific files or directories, 19
  small steps, 4
  smoke tests, 286, 299
  sort testing, 93